

Menu-Driven Program for Digit Operations

Aim: To write a menu-driven C program using switch-case to find the sum and product of the first and last digits of a positive integer, or to compare all its digits with a given digit k.

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu: (1) Find sum and product of first and last digits, (2) Compare the digits with a given digit k.
4. Input the user's choice.
5. Use switch-case according to the selected choice.
6. For choice 1, find the last digit using $n \% 10$.
7. Divide the number by 10 repeatedly until the first digit is obtained.
8. Calculate $\text{Sum} = \text{first} + \text{last}$ and $\text{Product} = \text{first} * \text{last}$, then display them.
9. For choice 2, input the digit k.
10. Extract each digit and compare it with k.
11. Increase the lesser, equal, or greater counter according to the comparison.
12. Display the three counts.
13. If the choice is invalid, display 'Invalid choice'.
14. Stop.

Test Case Table

Test	Number	Choice	k	Expected Output	Actual Output	Pass/Fail
1	12345	1	-	Sum = 6, Product = 5	Sum = 6, Product = 5	PASS
2	12345	2	3	Lesser = 2, Equal = 1, Greater = 2	Lesser = 2, Equal = 1, Greater = 2	PASS
3	5678	1	-	Sum = 13, Product = 40	Sum = 13, Product = 40	PASS
4	55525	2	5	Lesser = 1, Equal = 4, Greater = 0	Lesser = 1, Equal = 4, Greater = 0	PASS
5	9876	2	7	Lesser = 1, Equal = 1, Greater = 2	Lesser = 1, Equal = 1, Greater = 2	PASS
6	12345	3	-	Invalid choice	Invalid choice	PASS

C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, temp, digit, choice; int
```

```
    first, last, sum, product; int k;
```

```
    int lesser = 0, equal = 0, greater = 0;
```

```
    printf("Enter a positive integer: "); scanf("%d",  
    &n);
```

```
    if(n <= 0)
```

```

{
    printf("Please enter a positive integer."); return
    0;
}

```

```

printf("\n1. Find sum and product of first and last digits"); printf("\n2.
Compare digits with a given digit k"); printf("\nEnter your choice: ");
scanf("%d", &choice);

```

```

temp = n;

```

```

switch(choice)

```

```

{

```

```

    case 1:

```

```

        last = n % 10;

```

```

        while(temp >= 10)

```

```

            temp = temp / 10;

```

```

        first = temp;

```

```

        sum = first + last;

```

```

        product = first * last;

```

```

        printf("Sum = %d\n", sum);

```

```

        printf("Product = %d\n", product); break;

```

```

    case 2:

```

```

        printf("Enter the digit k: "); scanf("%d",
        &k);

```

```

        while(temp > 0)

```

```

        {

```

```

            digit = temp % 10;

```

```

            if(digit < k)

```

```

                lesser++;

```

```

            else if(digit == k)

```

```

                equal++;

```

```

            else

```

```

                greater++;

```

```

            temp = temp / 10;

```

```

        }

```

```

        printf("Count lesser = %d\n", lesser);

```

```

        printf("Count equal = %d\n", equal);

```

```

        printf("Count greater = %d\n", greater); break;

```

```

    default:

```

```

        printf("Invalid choice");

```

```

}

```

```

return 0;

```

```

}

```

Compilation and Execution

```
gcc program.c -o program
./program
```

Sample Test Run 1

```
Enter a positive integer: 12345 Enter
your choice: 1
Sum = 6
Product = 5
```

Sample Test Run 2

```
Enter a positive integer: 12345 Enter
your choice: 2
Enter the digit k: 3
Count lesser = 2
Count equal = 1
Count greater = 2
```

Result

The program was tested against the prepared use cases. The actual outputs matched the expected outputs for all listed test cases.

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